CM-ARM: Collaborative Model for Privacy-Preserving Association Rule Mining

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***Abstract*—Association analysis uncovers co-occurrence relationships in transactional data, including frequent itemsets and association rules, and supports a wide range of applications. However, privacy concerns often prevent direct data sharing. Differential privacy provides principled protection in both centralized and localized settings, yet existing methods for privacy-preserving association rule mining still face a practical trade-off between privacy, utility, and computational cost. This paper presents the Collaborative Model for Privacy-Preserving Association Rule Mining (CM-ARM), which coordinates a centralized step for selecting candidate items and determining a private truncation length with a localized step for identifying frequent itemsets and association rules. Extensive experiments on benchmark datasets and privacy budgets show that CM-ARM improves utility over prior localized approaches while substantially reducing end-to-end runtime, and it remains competitive in runtime efficiency when compared with traditional centralized baselines. To our knowledge, this is the first association rule mining framework that systematically combines centralized and localized differential privacy mechanisms to better balance privacy protection, analytical accuracy, and computational efficiency.**

***Index Terms*—Differential privacy, individual privacy, association analysis, association rule mining, frequent itemset mining.**

I. Introduction

ASSOCIATION analysis plays a crucial role in data analysis by identifying co-occurrences within transactional datasets, such as consumer consumption patterns. The advent of Big Data analytics has profoundly impacted consumer behavior and data management, raising significant privacy concerns. To address these concerns, the European Union implemented the General Data Protection Regulation (GDPR) in 2016, establishing comprehensive guidelines for data collection and processing [1].

To protect privacy, several techniques have been developed, including data anonymization and differential privacy (DP). Anonymization methods such as k-anonymity [2] and its variants provide a certain level of security but remain vulnerable to linkage attacks [3]. DP [4] offers robust privacy guarantees and is widely adopted in both academia and industry for purposes such as data publishing, data mining, and machine learning. Notable applications of DP include Google's RAPPOR in Chrome [5], Apple's use of DP in iOS 10 to track popular emojis and device performance [6], and Microsoft's implementation in healthcare data security [7].

While conventional DP efficiently manages privacy budgets, it typically requires data owners to send transaction information to a trusted curator. Local Differential Privacy (LDP) provides a solution by allowing data owners to anonymize their data locally before transmission, ensuring protection even from untrusted curators.

Various DP-compliant algorithms have been proposed for frequent itemset and association rule mining in centralized models, such as PrivBasis [8], FP-Growth [9], and DPARM [10], inspired by Apriori [11]. Google's RAPPOR algorithm [5] has laid the foundation for LDP-based pattern mining, with further advancements like Wang et al.'s SVIM and SVSM [12] under LDP.

Despite their advantages, both centralized and localized DP models have limitations. Centralized models depend on the trustworthiness of the curator, which is not always feasible. Localized models, on the other hand, increase the transmission and computational burden on data curators and users.

This paper makes the following contributions:

- We propose CM-ARM, a privacy-preserving association rule mining framework that combines centralized candidate selection and private truncation-length determination with localized mining of frequent itemsets and association rules.
- We evaluate CM-ARM against representative localized baselines (e.g., SVIM/SVSM) and traditional centralized baselines using standard utility metrics (Hits, Precision/Recall/F1-score, and support error) and end-to-end runtime, showing improved utility and substantially reduced runtime under the same privacy budgets.
- To our knowledge, this is the first association rule mining framework that integrates centralized and localized differential privacy mechanisms in a single pipeline to improve practical deployability.

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Across three benchmark datasets, CM-ARM achieves higher mining utility than a leading localized baseline under the same privacy budgets, and reduces end-to-end runtime by factors ranging from $6\times$ to $563\times$ (e.g., $563\times$ faster than SVSM on Kosarak and $463\times$ faster than Apriori on T10I4D100 K; see Section VI).

The structure of this paper is as follows: Section II introduces the preliminaries. Section III outlines the problem statement. Section IV provides a detailed description of the CM-ARM model. Privacy and utility analysis are discussed in Section V. Experimental results are presented in Section VI. Related literature is reviewed in Section VII, and the paper concludes in Section VIII.

II. Preliminaries

A. Association Rule Mining (ARM)

ARM identifies relationships between items in a dataset by analyzing transactions, where a transaction consists of items purchased together. The frequency of an item or itemset in transactions is measured by 'support'. For example, if milk appears in 60 out of 100 transactions, its support is 60%.

ARM reveals patterns like likely item co-purchases. For instance, supermarket data might show that milk and bread co-occur in 40% of transactions, with a confidence of 66.7%, indicating bread is bought in two-thirds of milk transactions. Similarly, diapers and beer, with a support and confidence of 40% and 66.7%, might indicate a common purchase pattern among fathers.

These insights demonstrate ARM's value in informing business strategies based on consumer behavior. Implementing ARM requires setting minimum support and confidence thresholds to filter significant associations. Established methods [11], [13] assign varied weights to items based on their transaction frequency and importance, reducing the number of itemsets needed for analysis and enhancing ARM's efficiency.

Definition 1 (Minimum Support). [11] For a given support relevance threshold $\rho \in [0, 1]$, the minimum allowable minimum item support (MIS) is $\lambda \in [0, 1]$. For an item j with support $sup_{\mathbf{T}}(j)$, the MIS of j is defined as:

$$\mathrm{MIS}(j) = \max\{\rho \times sup_{\mathbf{T}}(j), \lambda\}. \tag{1}$$

For an itemset X that is a subset of the item universe I, the minimum support is given by $\min_{j\in X}\{\mathrm{MIS}(j)\}$.

The MIS is constrained by the support relevance parameter ρ and the lowest allowable MIS threshold λ, both of which are within the range [0, 1]. When $\rho = 0$, the MIS remains fixed ($\mathrm{MIS}(j) = \lambda$), analogous to the traditional single-threshold frequent itemset mining scenario. A frequent itemset is defined as an itemset whose support exceeds the specified minimum threshold.

Definition 2 (Frequent Itemset [11]). Given an itemset $X \subseteq I$, and considering a dataset $\mathbf{T}$ with n transactions t, the support of X is defined as

$$sup_{\mathbf{T}}(X) = \frac{1}{n}|t : X \subseteq t|, \tag{2}$$

an itemset is deemed a frequent itemset if and only if

$$sup_{\mathbf{T}}(X) \geq \min_{j\in X} \mathrm{MIS}(j). \tag{3}$$

Relevant association rules can be derived from frequent itemsets, with these rules exhibiting both high frequency and confidence.

Definition 3 (Confident Association Rule [11]). Consider two itemsets $X, Y \subseteq I$, and an association rule $r : X \to Y$ present in the dataset $\mathbf{T}$. The support of rule r is defined as

$$sup_{\mathbf{T}}(r) = sup_{\mathbf{T}}(X \cup Y), \tag{4}$$

while the confidence of rule r is given by

$$conf_{\mathbf{T}}(r) = \frac{sup_{\mathbf{T}}(r)}{sup_{\mathbf{T}}(X)}. \tag{5}$$

B. Differential Privacy (DP)

DP ensuring the protection of individual privacy in data analysis can be classified into two privacy definitions: ε-DP and ε-LDP.

Definition 4 (ε-DP and ε-LDP [4]). A randomized mechanism $\mathcal{M}$ satisfies ε-DP if, for any neighboring datasets $\mathbf{D}$ and $\mathbf{D}'$, and for any query q with an output $\hat{q} \in \mathrm{range}(\mathcal{M})$:

$$\Pr[\mathcal{M}(\mathbf{D}, q) = \hat{q}] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(\mathbf{D}', q) = \hat{q}]. \tag{6}$$

ε-LDP, a specific case of ε-DP, applies the mechanism locally to each individual data point $v \in \mathbf{D}$ and $v' \in \mathbf{D}'$, maintaining the privacy of raw data:

$$\Pr\left[\mathcal{M}(v, q) = \hat{q}\right] \leq e^{\varepsilon} \cdot \Pr\left[\mathcal{M}(v', q) = \hat{q}\right]. \tag{7}$$

Achieving DP can be straightforwardly realized by adding Laplace noise to query results.

Definition 5 (Laplace Mechanisms for DP and LDP [4]). For a dataset $\mathbf{D}$, a query q, and a privacy budget ε, the Laplace mechanism $\mathcal{M}^{\mathrm{DP}}\mathrm{Lap}$ ensures ε-DP across the dataset:

$$\mathcal{M}^{\mathrm{DP}}\mathrm{Lap}(\mathbf{D}) = \mathrm{q}(\mathbf{D}) + \mathrm{Lap}\left(\frac{\Delta\mathrm{q}}{\varepsilon}\right), \tag{8}$$

and $\mathcal{M}^{\mathrm{LDP}}\mathrm{Lap}$ guarantees ε-LDP for each data entry v:

$$\mathcal{M}^{\mathrm{LDP}}\mathrm{Lap}(v) = \mathrm{q}(v) + \mathrm{Lap}\left(\frac{\Delta\mathrm{q}}{\varepsilon}\right), \tag{9}$$

with $\Delta\mathrm{q}$ representing the sensitivity of the query q, measured as the maximum change to q's output that could result from modifying a single data entry within $\mathbf{D}$.

Definition 6 (Global Sensitivity [4]). The global sensitivity of a real-value function query q is the distance between two query results from neighboring datasets:

$$\Delta\mathrm{q} = \max_{\mathbf{D},\mathbf{D}'} |\mathrm{q}(\mathbf{D}) - \mathrm{q}(\mathbf{D}')|. \tag{10}$$

The combination of several differentially private mechanisms still satisfies DP [14].

Definition 7 (Postprocessing Closure [14]). For any algorithm $\mathcal{A}$ and a randomized mechanism $\mathcal{M}$, if $\mathcal{M}$ satisfies ε-DP, then $\mathcal{A} \circ \mathcal{M}$ satisfies ε-DP.

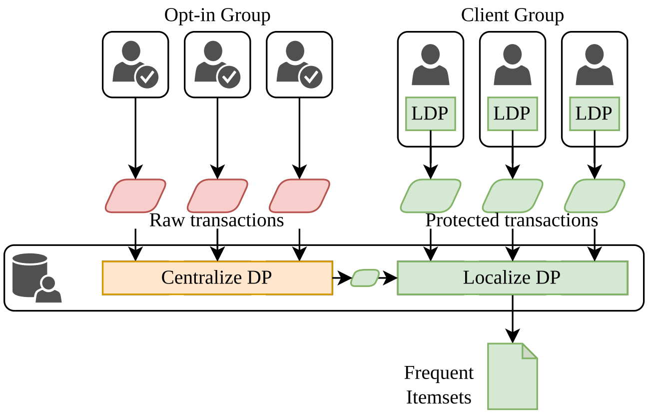

Fig. 1. A schematic representation of the proposed collaborative model for privacy-preserving association rule mining. This model integrates the benefits of both traditional DP and LDP.

Definition 8 (Sequential Composition [14]). For a dataset D, let each mechanism $\mathcal{M}i(D,q) = \hat{q}$ be an ε_i-DP algorithm for $i \in [k]$. For a set of such algorithms, $\mathcal{M}[k] = (\mathcal{M}_1 + \cdots + \mathcal{M}_k) = \prod_{i=1}^{k} \hat{q}_i$ is an $\left(\sum_{i=1}^{k} \varepsilon_i\right)$-DP algorithm.

Definition 9 (Parallel Composition [14]). For random partitions $D_1, \ldots, D_k$ of the dataset D, let each mechanism $\mathcal{M}i(D_i,q) = \hat{q}$ be an ε_i-DP algorithm for $i \in [k]$. For a set of such algorithms, $\mathcal{M}[k] = (\mathcal{M}_1 + \cdots + \mathcal{M}_k) = \prod_{i=1}^{k} \hat{q}_i$ is a $\left(\max_{j=1}^{k} \varepsilon_j\right)$-DP algorithm.

III. Problem Statement

A. Collaborative Model

DP provides stringent privacy guarantees, assuming attackers have maximal background knowledge [4] and minimizing re-identification risks [3]. There are two primary models: traditional DP and LDP. Traditional DP assumes a trusted curator who processes data privately, offering higher utility. LDP protects data directly on client devices, avoiding the need for a trusted third party but typically resulting in lower accuracy and efficiency. Both models ensure strict privacy.

However, stringent privacy guarantees often compromise practical utility. Assuming that all datasets must be entirely protected is unrealistic since the sensitivity of data within datasets varies. Different conceptions of privacy [15], [16] often lead to some users voluntarily adopting more lenient privacy models and sharing their data [17]. Based on this observation, modern research [18], [19] strives to define more relaxed forms of DP, which, while maintaining DP for sensitive data, allow the inclusion of unprotected non-sensitive data in computations [20], [21]. Avent et al. introduced the BLENDER algorithm, which combines traditional DP and LDP without a relaxed privacy definition [17].

Therefore, we propose a collaborative model (Fig. 1) for privacy-preserving association rule mining. This model merges the advantages of both traditional DP and LDP, enhancing the efficiency of time and the usage of the privacy budget in DP operations while ensuring privacy. Each user contributes a transaction; in our model, the user and transaction are considered equivalent. Users are categorized into two groups [17]: (1) Opt-in Group: Who trust the curator and consent to traditional DP. (2) Client Group: Users who prefer data privatization before sending it to the curator, protected by LDP.

Our model balances data protection and utility, assigning sensitive data to the Client Group and less sensitive data to the Opt-in Group, such as preliminary research, open datasets, and data from willing users.

B. Privacy-Preserving ARM Based on LDP

SVSM [12] is the first method for privacy-preserving ARM (PPARM) in the LDP setting. The SVSM protocol involves five stages to process transactions within the dataset $\mathbf{t} \subseteq T$. Initially, a single item is reported from each transaction to identify candidate items $S = \text{PSFO}_{\text{Adap}(1,\varepsilon)}(\mathbf{t})$, using either optimal local hashing (OLH) or optimal unary encoding (OUE). The second stage sets the transaction reporting size $L = \text{OLH}(\varepsilon)(\mathbf{t} \cap S)$. The third stage identifies frequent items using an adaptive Padding-and-Sampling-based Frequency Oracle (PSFO) $IS = \text{PSFO}_{\text{Adap}(L,\varepsilon)}(\mathbf{t})$. The final stages identify frequent itemsets similarly.

SVSM's first phase reduces communication costs by reporting only one item per transaction, but this can decrease utility, especially in small datasets. In contrast, Priv_OA [22] encodes nearly every item, improving accuracy but increasing communication costs.

Afrose et al. [22] proposed Priv_OA, an iterative LDP algorithm that encodes user data into binary arrays with random bit flips for privacy. While this increases accuracy, it also incurs high communication costs, making it suitable for small datasets only.

Table I compares several state-of-the-art association analysis methods. Considering both data curator and client costs is crucial. Our method CM-ARM balances communication costs and utility, enhancing accuracy under stringent privacy definitions. By integrating traditional DP, we are conservative in data collection and use, resulting in better time efficiency. The notation is detailed in Table II.

IV. CM-ARM Algorithm

To address the challenges outlined in Section III, we introduce the CM-ARM algorithm. This algorithm is designed to offer a more flexible, accurate, and fast PPARM method under strict DP privacy constraints.

A. Overview of the CM-ARM Algorithm

The CM-ARM algorithm's workflow is illustrated in Fig. 2. The algorithm is structured into five distinct phases, with centralized DP applied in the centralized plane during Phases 1 and 2 (highlighted in orange), and localized DP applied in the local plane during Phases 3 to 5. In Phase 1, we compute each item's MIS to establish the threshold for candidate selection. Phase 2 utilizes a centralized approach to identify candidate frequent items S and determine the truncation length θ. In Phase 3,

TABLE I
COMPARISON OF SEVERAL STATE-OF-THE-ART METHODS

Algorithms	Model	Pattern Type	Privacy Definition	Communication Cost
DPARM [10]	Centralized	Frequent itemsets	ε-DP	/
RAPPOR [5]	Distributed	Heavy Hitter	ε-LDP	/
SVSM [12]	Distributed	Frequent itemsets	ε-LDP	$O(\lvert\mathbf{T}'^{(1)}\rvert + \lvert\mathbf{T}'^{(2)}\rvert + \lvert S\rvert\lvert\mathbf{T}'^{(3)}\rvert + \lvert\mathbf{T}'^{(4)}\rvert + \lvert IS\rvert\lvert\mathbf{T}'^{(5)}\rvert)$
Priv_OA [22]	Distributed	Frequent itemsets	ε-LDP	$O(\lvert\lvert\mathbf{T}''^{(1)}\rvert\rvert + \lvert S\rvert\lvert\mathbf{T}''^{(2)}\rvert + \lvert IS_2\rvert\lvert\mathbf{T}''^{(3)}\rvert + \cdots + \lvert IS_m\rvert\lvert\mathbf{T}''^{(m)}\rvert)$
OUR	Collaborative	Frequent itemsets	ε-DP & ε-LDP	$O(\lvert\lvert\mathbf{T}^{(1)}\rvert\rvert + \lvert\lvert\mathbf{T}^{(2)}\rvert\rvert + \lvert S\rvert\lvert\mathbf{T}^{(3)}\rvert + \lvert\mathbf{T}^{(4)}\rvert + \lvert IS\rvert\lvert\mathbf{T}^{(5)}\rvert)$

Let $\mathbf{T}^{(i)}$ represent the quantity of data sets in Phase i. It is important to note that in different algorithms, each Phase is not identical, and the size of the data sets also varies. Therefore, we use $\mathbf{T}'^{(i)}$ and $\mathbf{T}''^{(i)}$ to distinguish between them. Communication cost is not reported for DPARM as it follows the centralized DP model with single data upload. It is also omitted for RAPPOR since it targets heavy hitter estimation rather than frequent itemset mining, making the communication pattern not directly comparable.

TABLE II
NOTATIONS

Notation	**Description**
$\mathbf{T}$	transactional dataset
θ	truncation length
p	truncation rate
ε	privacy budget
Δ	sensitivity
I	item universe
λ	lowest allowed MIS
ρ	support relevance
IS, FIS	frequent items / itemsets

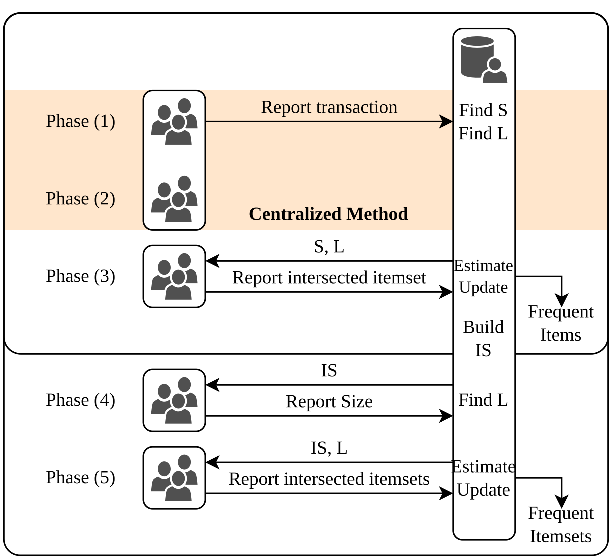

Fig. 2. Depiction of the CM-ARM algorithm's five-phase process. Centralized DP is utilized in phases 1 and 2, applied to the opt-in group for the identification of candidate frequent items and determination of reporting size. LDP is employed in phases 3, 4, and 5 for the client group, facilitating the identification of frequent items and itemsets.

the PSFO protocol [23] and an adaptive frequency oracle [12] are employed to ascertain frequent items IS. Phase 4 involves estimating the user-reported truncation length $\theta^{(4)}$ using the OLH method (as defined in Definition 12). Finally, in Phase 5, the same approach as in Phase 3 is used to derive frequent itemsets FIS.

Algorithm 1. Random Truncation.

Input: Data set $\mathbf{T}$, cardinality percentile p, privacy budget ε'

Output: Truncated data set $\mathbf{T}$, truncation length θ

1 **Function** RandomTruncation($\mathbf{T}, p, \varepsilon'$)**:**
2 $\quad \mathbf{T} = [\mathbf{T}_1, \ldots, \mathbf{T}_m]$;
$\quad$ /* $\mathbf{T}_j$ contains transactions with cardinality j */
3 $\quad \mathbf{d} = [d_1, \ldots, d_m] = [||\mathbf{T}_1||_1 + \mathrm{Lap}_1(\frac{1}{\varepsilon'}), \ldots, ||\mathbf{T}_m||_1 + \mathrm{Lap}_m(\frac{1}{\varepsilon'})]$;
4 $\quad \theta =$ the minimum j such that $\sum_{l=1}^{j} d_l \geqslant p(\mathcal{T}) \times ||\mathcal{T}||$;
5 $\quad$ **for** each transaction $t \in \mathbf{T}$ **do**
6 $\quad\quad$ **if** $|t| > \theta$ **then**
7 $\quad\quad\quad t =$ randomly sample θ items from t;
8 $\quad\quad$ **end**
9 $\quad$ **end**
10 $\quad$ **return** $\mathbf{T}, \theta$
11 **end**

Prior to detailing the main algorithm, we introduce two fundamental techniques: random truncation and noisy count, in the following subsections.

B. Random Truncation

The process of random truncation truncates each transaction in the dataset T to a fixed maximum length θ. This process controls computational complexity by reducing the dataset's dimension, thereby lowering time consumption and communication costs.

Algorithm IV-B outlines the process of random truncation. In Algorithm IV-B, lines 2–3 partition $\mathcal{T}$ by transaction length and privately release a *noisy length histogram*. Specifically, for each length $j \in [1, m]$, the query is the bin count $q_j(\mathcal{T}) = ||\mathcal{T}_j||_1$ (i.e., the number of transactions whose cardinality equals j), which is the target of noise addition. Since modifying one transaction affects the count of any *single* length bin by at most 1, each bin-count query has sensitivity $\Delta q = 1$; thus each bin is perturbed as $d_j = q_j(\mathcal{T}) + \mathrm{Lap}(1/\varepsilon')$ (line 3), where ε' is the privacy budget allocated to the truncation step. In line 4, the truncation length θ is determined from the noisy cumulative counts as the smallest value such that $\sum_{l=1}^{\theta} d_l \geq p(\mathcal{T}) \cdot ||\mathcal{T}||$,

Algorithm 2. Noisy Count.

Input: Data set $\mathbf{T}$, candidate itemsets $\mathbf{G}$, sensitivity Δ', privacy budget ε'
Output: noisy counted itemsets $\mathbf{G}$

1 **Function** $\mathrm{NoisyCount}(\mathbf{T}, G, \Delta', \varepsilon')$**:**
2   **for** each transaction $t \in \mathbf{T}$ **do**
3   **for** each candidate itemset $g \in \mathbf{G}$ **do**
4   **if** $g \subseteq t$ **then**
5         $g.count++$
6   **end**
7   **end**
8   **end**
9   **for** each counted itemset $g \in \mathbf{G}$ **do**
10     $g.sup = g.count/|\mathbf{T}| + \mathrm{Lap}(\frac{\Delta}{\varepsilon})$;
11   **end**
12   **return G**
13 **end**

i.e., covering the shortest $p(\mathcal{T})$ fraction of transactions under the noisy histogram. Finally, lines 5–9 enforce this bound by truncating any transaction with $|t| > \theta$ via uniformly sampling θ items without replacement, yielding a dataset with maximum transaction length θ while ensuring ε'-DP for releasing the noisy histogram (and thus selecting θ).

In all experiments we use $p(\mathcal{T}) = 0.85$ for consistency; optimizing $p(\mathcal{T})$ is orthogonal and can be done via the utility-driven search in [24].

Additionally, random truncation influences the communication costs of CM-ARM. This highlights the connection between the truncation lengths in the centralized plane (referenced in Algorithm IV-B) and the report sizes in the local plane. In Phases 1 and 2, each user reports at most $\theta^{(1)}$ and $\theta^{(2)}$ items. However, the adoption of a centralized model enhances the efficiency of these phases, allowing them to operate effectively with a smaller portion of the dataset.

We have proven that the process of random truncation satisfies ε-DP as demonstrated in Theorem 2.

C. Noisy Count

The process of noisy count (in Algorithm IV-C) calculates the support for itemsets by adding noise to achieve DP protection. First, in lines 2 to 8, the algorithm iterates through each candidate $g.sup$ to count their respective supports within the transactions $\mathcal{T}$, considering whether a candidate itemset $g \in \mathcal{G}$ is present in a transaction $t \in \mathcal{T}$. Subsequently, in lines 9 to 11, Laplace noise $\mathrm{Lap}(\frac{\Delta}{\varepsilon})$ is introduced to these supports. The process of noisy count satisfying ε-DP has demonstrated in Theorem 3.

D. Main Algorithm

We are now prepared to explain our main algorithm, CM-ARM, which is presented in Algorithm 3. The CM-ARM algorithm is divided into five distinct phases, each represented by separate code blocks. Superscripts (e.g., $T^{(i)}$ for phase i) are used to distinguish regional variables across phases. The

Algorithm 3. CM-ARM Algorithm.

Input: Data set $\mathbf{T}$, privacy budget ε, support relevance ρ, Lowest allowable MIS λ
Output: Frequent itemsets IS

1 **Phase** 1**:**
2   $\varepsilon_T^{(1)}, \varepsilon_C^{(1)} = \varepsilon$;
3   $T^{(1)}, \theta^{(1)} = \mathrm{RandomTruncation}(T^{(1)}, p, \varepsilon_T^{(1)})$;
4   $\Delta^{(1)} = \theta^{(1)}/|\mathbf{T}^{(1)}|$;
5   $I = \mathrm{NoisyCount}(\mathbf{T}^{(1)}, I, \Delta^{(1)}, \varepsilon_C^{(1)})$;
6   **for** each item $i \in I$ **do**
7     $i.\mathrm{MIS} = \max\{\rho \times i.sup, \lambda\}$;
8   **end**
9   $I = \mathrm{Sort}(I)$;   // Sort by $i.\mathrm{MIS}$
10   **return** I
11 **end**
12 **Phase** 2**:**
13   $\varepsilon_T^{(2)}, \varepsilon_C^{(2)} = \varepsilon$;
14   $T^{(2)}, \theta^{(2)} = \mathrm{RandomTruncation}(T^{(2)}, p, \varepsilon_T^{(2)})$;
15   $\Delta^{(2)} = \theta^{(2)}/|\mathbf{T}^{(2)}|$;
16   $I = \mathrm{NoisyCount}(\mathbf{T}^{(2)}, I, \Delta^{(2)}, \varepsilon_C^{(2)})$;
17   $\mathbf{C} = \varnothing$;
18   $\gamma = \min\{i.\mathrm{MIS}\}$ where $i.sup \geqslant i.\mathrm{MIS}, i \in I$;
19   **for** each item $i \in I$ **do**
20   **if** $i.sup \geqslant \gamma$ **then** $\mathbf{C} = \mathbf{C} \cup \{\{i\}\}$;
21   **end**
22   $S = \varnothing$;
23   **for** each itemset $c \in \mathbf{C}$ **do**
24     $c.MS = \min_{i \in c}\{i.\mathrm{MIS}\}$;
25     $c.sup = \min_{i \in c}\{i.sup\}$;
26   **if** $c.sup \geqslant c.MS$ **then** $S = S \cup c$;
27   **end**
28   **return** $S, \theta^{(2)}$
29 **end**
30 **Phase** 3**:**
31   $\mathrm{IS} = \mathrm{PSFO}_{\mathrm{Adap}(\theta^{(2)}, \varepsilon)}(\mathbf{t} \cap S), \mathbf{t} \subseteq T^{(3)}, |\mathrm{IS}| = 2k$;
32 **end**
33 **Phase** 4**:**
34   $\mathbf{vs}^{(4)} = \{\mathbf{x} : \mathbf{x} \in \mathrm{IS}, \mathbf{x} \subseteq \mathbf{t}\}, \mathbf{t} \subseteq T^{(4)}$;
35   $\theta^{(4)} = \mathrm{OLH}_{(\varepsilon)}(|\mathbf{vs^{(4)}}|)$;
36 **end**
37 **Phase** 5**:**
38   $\mathbf{vs}^{(5)} = \{\mathbf{x} : \mathbf{x} \in \mathrm{IS}, \mathbf{x} \subseteq \mathbf{t}\}, \mathbf{t} \subseteq T^{(5)}$;
39   $\mathrm{FIS} = \mathrm{PSFO}_{\mathrm{Adap}(\theta^{(4)}, \varepsilon)}(\mathbf{vs}^{(5)})$;
40 **end**

algorithm begins in the centralized plane: Phase 1 (lines 1 to 11) involves discovering the MIS, and Phase 2 (lines 12 to 29) focuses on identifying candidate frequent items and truncation lengths. Transitioning to the localized plane, Phase 3 (lines 30 to 32) uncovers frequent items, Phase 4 (lines 33 to 36) determines the reported length for frequent itemsets, and Phase 5 (lines 37 to 40) finalizes the identification of frequent itemsets.

Within Phases 1 and 2 of the centralized plane, the primary objective is to identify candidate itemsets. Specifically, lines 2 to 5 and 13 to 16 focus on calculating noisy counts. In line 2, the privacy budget ε is allocated into two segments: ε_T for truncation and ε_C for noisy counts. Subsequently, in line 3, transactions are randomly truncated to the appropriate size, yielding the truncation length θ via Algorithm IV-B. The sensitivity $\Delta = \frac{\theta}{|T|}$ is computed in line 4, followed by the calculation of itemset supports via Algorithm IV-C in line 5, safeguarded by the Laplace mechanism.

Phase 1 (lines 6 to 8) involves computing the MIS i.MIS for each item $i \in I$, as defined in Definition 1. This threshold facilitates the exploration of low-frequency and high-confidence correlations, and the items are subsequently sorted by their MIS in line 9. A frequent item i is equivalent to a frequent itemset containing only one item $\{i\}$. In Phase 2 (lines 17 to 21), the domain C is refined to include items whose support i.sup exceeds the minimum MIS, $\min i$.MIS, such that i.sup $\geq i$.MIS. Sorting items by their MIS allows for the derivation of γ (line 18), determined from lines 19 to 21 upon the first yield. Consequently, a single pass suffices to acquire C. Lines 22 to 27 then identify candidate items (or itemsets containing a single item) S, including itemsets with a minimum support surpassing their individual support.

In the localized phases, the PSFO protocol [23] and an adaptive frequency oracle [12] are employed. Phase 3 (line 31) utilizes the adaptive PSFO method to identify frequent items IS consisting of 2k items. The specifics of the PSFO method are detailed in Definition 10. This method allows the utilization of two frequency oracles: Generalized Randomized Response (GRR) in Definition 11 or OLH in Definition 12. The adaptive method for selecting a frequency oracle is elucidated in Definition 13. Phase 4 involves estimating the user-reported truncation length $\theta^{(4)}$ by evaluating the frequency of frequent items, achieved using OLH. Phase 5, which focuses on identifying frequent itemsets FIS, employs a methodology similar to that of Phase 3.

Definition 10. The PSFO protocol [12] uses the frequency oracle protocol to transmit elements while preserving privacy. Consider a pair of algorithms $\langle \Psi, \Phi \rangle$ for the curator and users. Given a truncation length θ and a frequency oracle algorithm FO with a privacy budget ε, we define:

$$\text{PSFO}(\theta, \text{FO}, \varepsilon) := \langle \Psi_{\text{FO}(\varepsilon)}(\text{PS}(\cdot)), \Phi_{\text{FO}(\varepsilon)}(\cdot) \cdot \theta \rangle. \tag{11}$$

Two types of frequency oracle protocol are used.

Definition 11. GRR [25] gives the value with probability p. For domain D, a privacy budget of ε, and a value v reported as x, we have

$$\forall x \in D, \Pr[\Psi_{\text{GRR}(\varepsilon)}(v) = x] = \begin{cases} p = \frac{e^\varepsilon}{e^\varepsilon + d - 1} & \text{if} \quad x = v \\ q = \frac{1}{e^\varepsilon + d - 1} & \text{if} \quad x \neq v \end{cases}. \tag{12}$$

With reported value $x' \in D$, when the frequency can be estimated as v', we have

$$\Phi_{\text{GRR}(\varepsilon)}(v') := \frac{\text{count}(\mathrm{x}') - nq}{p - q}. \tag{13}$$

Because $p/q = e^\varepsilon$, GRR satisfies ε-LDP.

Definition 12. OLH [26] maps the large domain d onto a small domain $g = \lceil e^\varepsilon + 1 \rceil$ using the hash function $\mathcal{H}$. We then report the hashed value with a random response as follows.

$$\Psi_{\text{OLH}(\varepsilon)}(v) := \langle \mathcal{H}, \Psi_{\text{GRR}(\varepsilon)}(\mathcal{H}(v)) \rangle. \tag{14}$$

With reported value x', when the frequency can be estimated as v', we have

$$\Phi_{\text{OLH}(\varepsilon)}(v') := \frac{\text{count}(\mathrm{x}') - n/g}{p - 1/g}. \tag{15}$$

In each localized phase, we define the domain size $d := |\mathcal{D}|$ as the cardinality of the current candidate set $\mathcal{D}$ (items/itemsets) to which the frequency oracle is applied. At the beginning of that phase, the curator computes d and selects the oracle using the phase-specific (ε, θ): if $d < e^\varepsilon \theta(4\theta - 1) + 1$ (small domain), PSFO adopts GRR with the amplified internal budget $\varepsilon' = \ln(\theta(e^\varepsilon - 1) + 1)$; otherwise (large domain), PSFO adopts OLH with ε. In our pipeline, GRR is typically used after candidate pruning yields a small $\mathcal{D}$, while OLH is used when $\mathcal{D}$ is large (e.g., itemset stages with many candidates).

Definition 13. For an adaptive frequency oracle [12] with a privacy budget of ε and a truncation length θ, we have

$$\begin{aligned} &\text{Adap}(\varepsilon, \theta) \\ &:= \begin{cases} \text{GRR}(\ln(\theta(e^\varepsilon - 1) + 1)) & \text{if } d < e^\varepsilon \theta(4\theta - 1) + 1, \\ \text{OLH}(\varepsilon) & \text{otherwise} \end{cases} \end{aligned} \tag{16}$$

In subsequent sections, we conduct a rigorous analysis of the utility and privacy of the CM-ARM algorithm.

V. System Analysis

To analyze our system, we estimate the data utility in terms of error variance and prove that our CM-ARM algorithm satisfies ε-DP.

A. Utility Analysis

Theorem 1. The utility for our CM-ARM in terms of error variance can be expressed as:

$$\text{Var}(\bar{f}_x) = \frac{1}{n^2} \sum_{i=\theta+1}^{m} (i - \theta)^2 \text{Var}(T_i), \tag{17}$$

where $\bar{f}_x$, n, m, and θ denote truncated items frequency, total item count, transactions, and truncation length, respectively.

Proof. Initially, we represent the total item count as:

$$n = \sum_{i=1}^{\theta} iT_i + \sum_{i=\theta+1}^{m} iT_i = \sum_{i=1}^{m} iT_i. \tag{18}$$

After performing Algorithm IV-B, we can obtain the truncated data set T with truncation length θ:

$$\sum_{i=1}^{\theta} iT_i + \sum_{i=\theta+1}^{m} \theta T_i = n - \sum_{i=\theta+1}^{m} (i - \theta)T_i, \tag{19}$$

where the truncated item count is specified by $\sum_{i=\theta+1}^{m} (i - \theta)T_i$ and denoted as c for ease of expression. Then, the probability of truncated items $\mathbb{P}(c)$ is calculated as:

$$\mathbb{P}(c) = \frac{1}{n}\left(\sum_{i=j+1}^{m}(i-\theta)T_i\right). \tag{20}$$

Subsequently, the truncated items frequency can be defined as:

$$\bar{f}_x = f_x - \frac{1}{n}\sum_{i=\theta+1}^{m}(i-\theta)T_i. \tag{21}$$

According to the above equations, we can obtain the truncated item count and support, which are expressed as follows.

$$x.\bar{count} = (n-c)\bar{f}_x, \tilde{}x.\bar{s}up = \frac{(n-c)\bar{f}_x}{|T|}. \tag{22}$$

Applying DP to truncated item support, the differentially private support can be denoted as:

$$x.\bar{\bar{s}}up = \frac{(n-c)\bar{f}_x}{|T|} + \mathrm{Lap}(\frac{\Delta}{\varepsilon}). \tag{23}$$

According to the differentially private support obtained from (23), we then define the error variance σ^2 of the utility as follows.

$$\sigma^2 = \mathrm{Var}(x.\bar{\bar{s}}up) = \mathrm{Var}(x.\bar{s}up) + 2\left(\frac{\Delta}{\varepsilon}\right)^2, \tag{24}$$

in which $2(\frac{\Delta}{\varepsilon})^2$ from the Laplace noise introduced to ensure differential privacy, and $\mathrm{Var}(x.\bar{s}up)$ $(= \frac{(n-c)^2}{|T|^2}\mathrm{Var}(\bar{f}_x))$ represents the variance due to truncation.

Finally, $\mathrm{Var}(\bar{f}_x)$ can be derived from the variance of the truncated item frequency and written as:

$$\mathrm{Var}(\bar{f}_x) = \frac{1}{n^2}\sum_{i=\theta+1}^{m}(i-\theta)^2\,\mathrm{Var}(T_i). \tag{25}$$

□

B. Privacy Analysis

In our collaborative model, users are divided into an opt-in group for centralized DP and a client group for LDP. Because LDP is a particular case of centralized DP, we can prove that our collaborative model CM-ARM satisfies the definition of ε-DP.

In phases 1 and 2, we add noise to calculate the cardinality distribution in the random truncation Algorithm IV-B and compute the supports in the noisy counting Algorithm IV-C.

Theorem 2. The process of random truncation satisfies ε-DP.

Proof. To demonstrate ε-DP, consider two neighboring datasets $\mathbf{S}$ and $\mathbf{S}'$ differing by at most one element. We need to prove that the probability ratio for any output $\hat{q}$ of our mechanism $\mathcal{M}$ conforms to the DP criterion:

$$\frac{\Pr[\mathcal{M}(\mathbf{S}) = \hat{q}]}{\Pr[\mathcal{M}(\mathbf{S}') = \hat{q}]} \le e^{\varepsilon}. \tag{26}$$

Our mechanism $\mathcal{M}$ outputs the protected item supports $x.\bar{\bar{s}}up$, as defined in (17). The randomization in $\mathcal{M}$ comes from the addition of Laplacian noise $\mathrm{Lap}(\frac{\Delta}{\varepsilon})$ to the truncated item supports $x.\bar{s}up$.

Given the sensitivity Δ of $x.\bar{s}up$ in the context of adding or removing an item is 1 (the maximum change in count by adding or removing an element), and the scale of the Laplace distribution is set to $\frac{\Delta}{\varepsilon}$, the probability density function of the Laplace mechanism at value v is given by:

$$f(v|\mu) = \frac{\varepsilon}{2\Delta}\exp\left(-\frac{\varepsilon|v-\mu|}{\Delta}\right), \tag{27}$$

where μ is the real output without noise.

For any output $\hat{q}$ and for the two neighboring datasets $\mathbf{S}$ and $\mathbf{S}'$, the ratio of their probabilities under our mechanism $\mathcal{M}$ can be bounded as follows:

$$\begin{aligned}\frac{\Pr[\mathcal{M}(\mathbf{S}) = \hat{q}]}{\Pr[\mathcal{M}(\mathbf{S}') = \hat{q}]} &= \frac{f(\hat{q}|x.\bar{s}up)}{f(\hat{q}|x.\bar{s}up')}\\ &= \exp\left(-\varepsilon\left|\frac{\hat{q}-x.\bar{s}up}{\Delta}\right| + \varepsilon\left|\frac{\hat{q}-x.\bar{s}up'}{\Delta}\right|\right)\\ &\le \exp(\varepsilon).\end{aligned}$$

It is evident that the process of random truncation satisfies the ε-DP criterion. □

Theorem 3. The process of noisy count satisfies ε-DP.

Proof. The fact that the process of noisy count satisfies ε-DP can be demonstrated by leveraging the proof of Theorem 2. In this case, the sensitivity Δ is replaced by $\theta/|T|$, reflecting the specific characteristics of the noisy count process. By applying the same reasoning and substituting the appropriate sensitivity parameter, we can conclude that the noisy count process adheres to the ε-DP criterion. □

Theorem 4. The CM-ARM model satisfies ε-DP.

Proof. The collaborative model, denoted as $\mathcal{M}$, can be decomposed into two distinct submodels: a centralized submodel, $\mathcal{M}_C$, which satisfies ε-DP, and a localized submodel, $\mathcal{M}_L$, which adheres to ε-LDP constraints.

Transactions $\mathcal{T}$ can be split into multiple disjoint sets, which can be simplified into $\mathcal{T}_C$ and $\mathcal{T}_L$ for centralized $\mathcal{M}_C(\mathcal{T}_C)$ and localized $\mathcal{M}_L(\mathcal{T}_L)$, respectively, where $\mathcal{T}_C \cap \mathcal{T}_L = \varnothing$ and $\mathcal{T}_C \cup \mathcal{T}_L = \mathcal{T}$.

Suppose the centralized submodel $\mathcal{M}_C$ satisfies ε_C-DP and the localized submodel $\mathcal{M}_L$ satisfies ε_L-LDP according to Definition 4. Since ε_L-LDP implies ε_L-DP as defined in [14], the collaborative model $\mathcal{M}$ satisfies $\max(\varepsilon_C, \varepsilon_L)$-DP.

Consequently, the collaborative model $\mathcal{M} = [\mathcal{M}_C, \mathcal{M}_L]$ satisfies $\max(\varepsilon_C, \varepsilon_L)$-DP in accordance with Definition 9, and can be intuitively simplified as ε-DP.

During phases 1 and 2, for each phase with a data set $\mathbf{T}^{(i)}, i \in [1, 2]$, a truncation algorithm $\mathcal{M}_T(\mathbf{T}^{(i)})$ is applied and $\varepsilon_T^{(i)}$-DP is satisfied; the proofs are in Theorem 2. The noisy counting algorithm $\mathcal{M}_C(\mathbf{T}^{(i)})$ satisfies $\varepsilon_C^{(i)}$-DP; $\varepsilon_T^{(i)} + \varepsilon_C^{(i)} = \varepsilon^{(i)}$. According to Definition 8, phase i satisfies $\varepsilon^{(i)}$.

For each phase $\mathbf{T}^{(i)}, i \in [1, 5]$ is a random sample of data set $\mathbf{T}$. Phases 1 and 2 satisfy ε-DP, and phases 3 to 5 satisfy ε-LDP according to Definition 12. Therefore, the CM-ARM algorithm satisfies ε-DP. □

VI. Experimental Results

We improved the performance of our local-setting data aggregation model by integrating elements from a centralized setting.

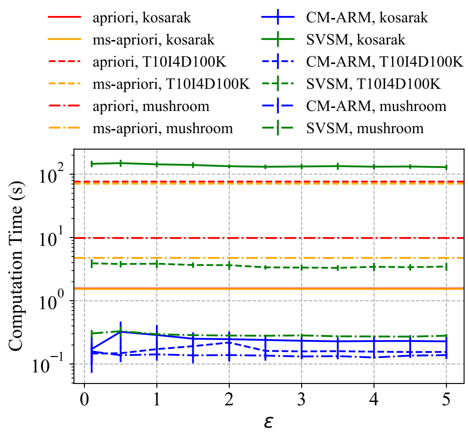

Fig. 3. Computation time under varying privacy budgets ϵ on Kosarak, T10I4D100 K, and Mushroom datasets. CM-ARM consistently achieves substantial speedups compared with SVSM and the non-private Apriori/MS-Apriori baselines. Notably, the computational cost of the ground-truth algorithms is independent of ϵ.

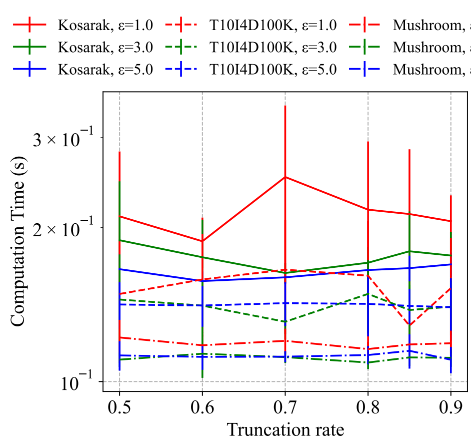

Fig. 4. Impact of truncation rate p on computation time under different privacy budgets ϵ. Results are reported on Kosarak, T10I4D100 K, and Mushroom datasets. CM-ARM maintains stable computational cost across varying truncation rates.

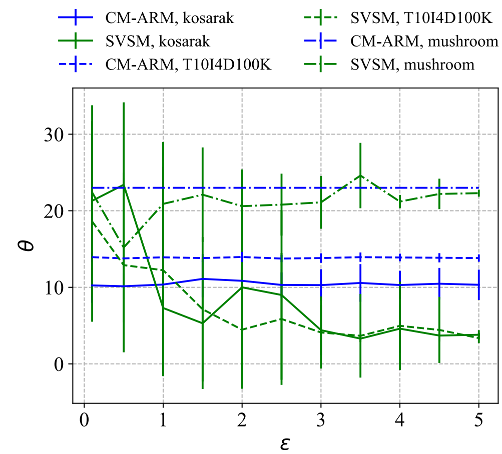

Fig. 5. Variation of truncation length θ under different privacy budgets ε on Kosarak, T10I4D100 K, and Mushroom datasets. CM-ARM exhibits relatively stable truncation lengths across varying privacy budgets, while SVSM shows larger fluctuations, particularly under smaller ε.

In this section, we benchmarked CM-ARM against SVSM [12], a similar local-setting ARM algorithm, and compared it using frequent items derived from the Apriori and MS-Apriori algorithms. Notably, the MS-Apriori algorithm was set as the ground truth in our experiments.

Our implementation and experiment scripts are available at https://github.com/undecV/CMARM-Experiment. For the SVSM baseline, we directly adopt the publicly available implementation at https://github.com/vvv214/LDP_Protocols.

We evaluated the algorithms based on the number of hits, F1-score, normalized cumulative rank (NCR) [30], and squared error. The NCR, indicating top-k ranking accuracy, is calculated as follows:

$$\mathrm{NCR} = \frac{\sum_{s\in S} q(s)}{\sum_{s\in S_g} q(s)}. \tag{28}$$

The quality function q assigns scores to the top k values, with the highest value receiving k points, the second highest $k-1$ points, and so on, down to 1 point; all other values receive a score of 0. The NCR normalizes these scores between 0 and 1 by dividing the total score by the maximum possible score, $k(k+1)/2$ [12]. Additionally, we used squared error to measure the discrepancy between the identified frequent itemsets and the ground truths:

$$\mathrm{Var} = \frac{1}{S\cap S_g}\sum_{s\in S\cap S_g} (f_{s_g} - \Phi(s))^2. \tag{29}$$

TABLE III
CHARACTERISTICS OF DATA SETS

Dataset	$\lvert T\rvert$	$\lvert I\rvert$	avg$\lvert t\rvert$	max $\lvert t\rvert$
Kosarak [27]	990002	41270	8	2498
T10I4D100K [28]	100000	870	10	29
Mushroom [29]	8124	119	23	23

The datasets were obtained from the Frequent Itemset Mining Implementations Repository [31], including Kosarak [27], T10I4D100K [28], and Mushroom [29]. Kosarak and T10I4D100 K were evaluated on 20,000 sampled transactions for controlled comparison, whereas the entire Mushroom dataset (8,124 transactions) was used due to its smaller size.

For centralized DP, 20% of the evaluated transactions were utilized (2% in Phase 1 and 18% in Phase 2).

All experiments were implemented in Python 3 and executed on a Windows 11 workstation with an Intel Core i7-9750H CPU (2.6 GHz) and 32 GB RAM. Each configuration was repeated multiple times, and averaged results are reported.

Table III summarizes the dataset characteristics.

A. Computation Time Under Varying ε and Truncation Rate

Fig. 3 shows the runtime under varying privacy budgets ε. Runtime is largely insensitive to ε, indicating that the privacy budget affects noise magnitude rather than computational complexity. Minor fluctuations in the sub-second regime are attributable to operating system scheduling noise and do not affect overall trends.

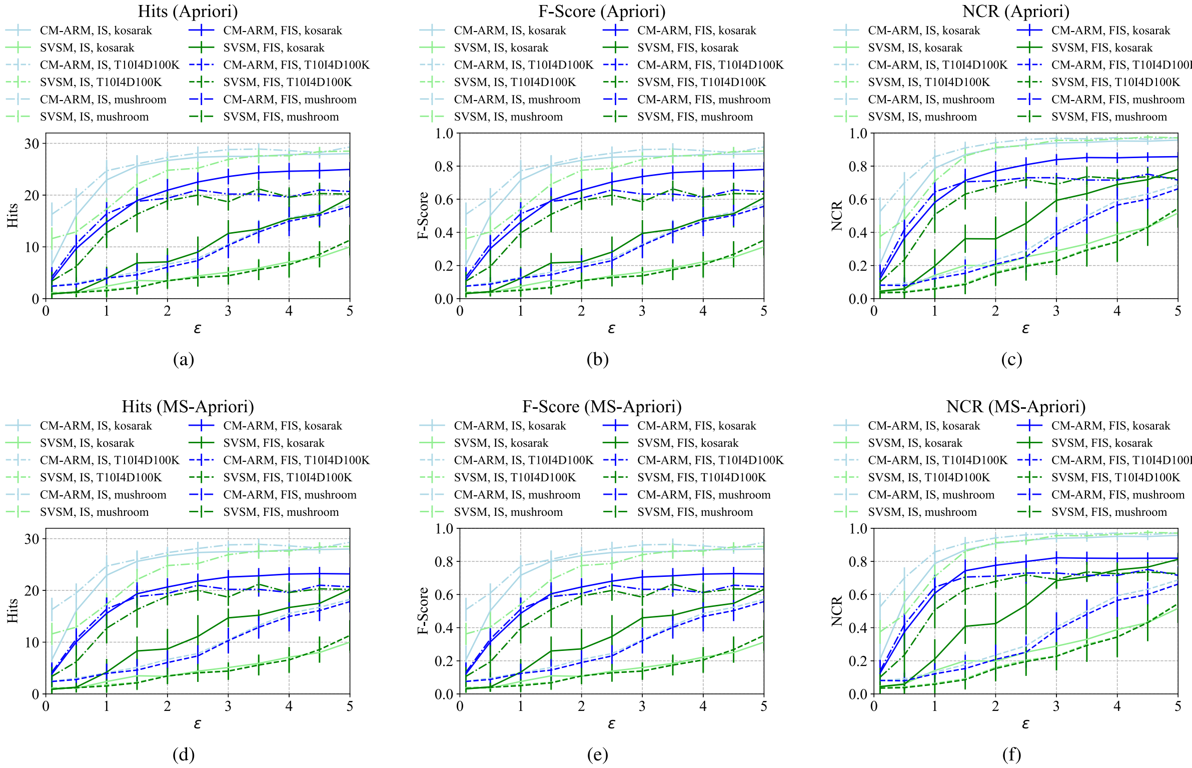

Fig. 6. Utility comparison under varying privacy budgets ε based on Apriori and MS-Apriori. Performance is evaluated using Hits, F-score, and NCR for both IS and FIS on the Kosarak, T10I4D100 K, and Mushroom datasets.

CM-ARM consistently achieves the lowest runtime across all datasets. On Kosarak, CM-ARM runs in 0.24 s on average, compared to 136.82 s for SVSM and 1.56 s for Apriori, achieving speedups of up to $563\times$ and $6\times$, respectively. On T10I4D100K, CM-ARM (0.17 s) outperforms SVSM (3.56 s) and Apriori (76.56 s), yielding $22\times$ and $463\times$ improvements. On the Mushroom dataset, CM-ARM ($\sim$0.2 s) achieves approximately $15\times$ speedup over SVSM and nearly $50\times$ over Apriori. Overall, CM-ARM provides speed improvements ranging from $6\times$ to over $563\times$ across datasets.

We further evaluate the impact of Truncation Rate on runtime. Since truncation length is determined from this rate and varies across datasets, we report results using Truncation Rate for unified comparison. As shown in Fig. 4, runtime remains stable across different Truncation Rates and privacy budgets, indicating that computational cost is dominated by candidate processing rather than the specific truncation choice. While SVSM scales with the item universe size and Apriori with dataset size and candidate growth, CM-ARM maintains substantially lower and stable runtime due to its hybrid design and efficient truncation mechanism.

B. Truncation Length Across Different Values of ε

Fig. 5 shows the truncation length θ under different privacy budgets ε. SVSM exhibits unstable θ under tight privacy budgets due to reporting only one item per transaction in Phase 1, whereas CM-ARM maintains stable truncation behavior across all ε values.

In SVSM, Phases 2 and 4 report up to the top 90% of candidate frequencies per transaction depending on protected candidates, but Phase 1 extracts minimal information. This imbalance leads to unstable truncation estimation. Choosing θ is critical: overly small values miss frequent itemsets, while overly large values increase communication cost and privacy budget usage.

In CM-ARM (Algorithm IV-B), θ is determined from the top $p\%$ of the length distribution in Phase 2, providing consistent truncation control. On the Mushroom dataset, where transaction length is fixed, CM-ARM produces nearly identical θ across ε, while SVSM still shows noticeable fluctuations.

C. Utility Under Varying ε and Truncation Rate

Fig. 6 evaluates utility under varying privacy budgets ε in terms of Hits, F-score, and NCR for IS and FIS on the Kosarak, T10I4D100 K, and Mushroom datasets. For fair comparison, the top-32 frequent items and itemsets are extracted from ground truth results based on Apriori and MS-Apriori.

Across all datasets, utility increases with ε. CM-ARM consistently outperforms SVSM under low and moderate privacy budgets. On Kosarak with $\varepsilon = 0.5$, CM-ARM achieves NCR values $7.38\times$ and $5.54\times$ higher than SVSM for IS and FIS, respectively. On T10I4D100K, CM-ARM remains superior even under tight privacy budgets, although performance becomes less stable for $\varepsilon < 2$.

For the Mushroom dataset (a low-sample, relatively high-dimensional dataset), both methods exhibit mild fluctuations at

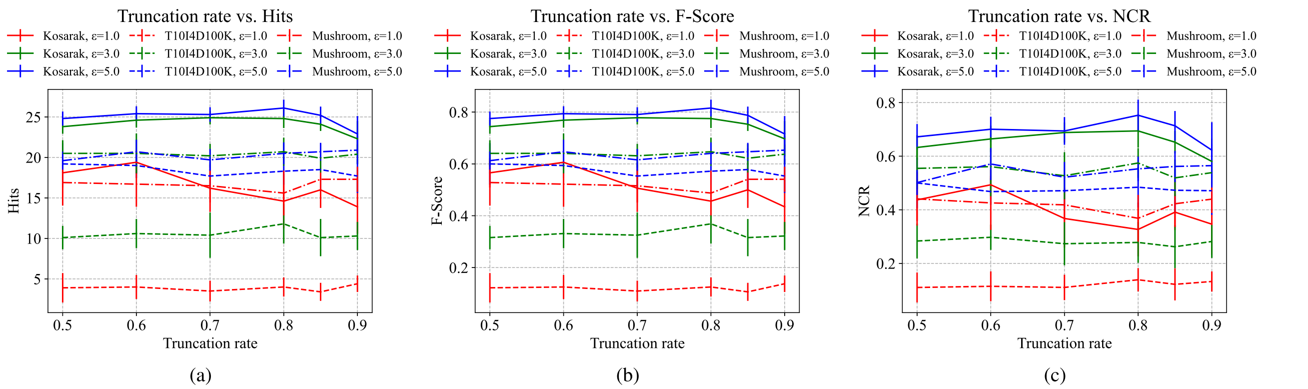

Fig. 7. Impact of Truncation Rate p on utility under different privacy budgets ε. Results are reported using Hits, F-score, and NCR for IS and FIS on the Kosarak, T10I4D100 K, and Mushroom datasets.

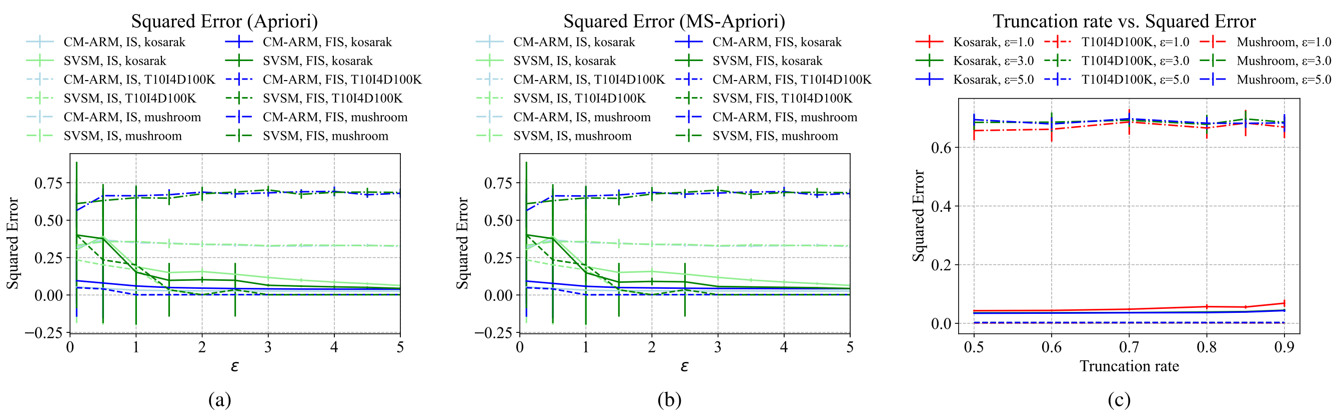

Fig. 8. Squared error under varying privacy budgets ε and Truncation Rates. Results are reported for IS and FIS based on Apriori and MS-Apriori.

higher ε, yet overall utility still increases with ε. Under low privacy budgets, CM-ARM maintains higher Hits, F-score, and NCR than SVSM.

Squared error results are shown in Fig. 8. Squared error decreases and stabilizes as ε increases. CM-ARM achieves lower error on Kosarak and T10I4D100 K, while on Mushroom both methods yield comparable error levels, with CM-ARM exhibiting slightly smaller variance.

Fig. 7 further evaluates the impact of Truncation Rate on utility. As observed in the figure, utility differs under various Truncation Rates, with moderate settings (around 0.8–0.85) yielding relatively stable and competitive performance. Since the best-performing Truncation Rate varies across datasets, a unified setting of 0.85 is adopted throughout the experiments for consistency.

As shown in Fig. 8, squared error is largely insensitive to Truncation Rate but remains dataset-dependent, with the Mushroom dataset exhibiting consistently higher error than Kosarak and T10I4D100 K.

VII. Related Work

ARM was introduced in [32] and is commonly realized through FIM using Apriori [11] and FP-Growth [33]. Mining transactional data, however, can expose sensitive information, motivating privacy-preserving ARM methods across multiple directions [34]; in this work, we focus on differential privacy.

Central DP for FIM/ARM: Under the trusted-curator model, DP mechanisms are applied after collecting raw transactions, typically enabling accurate noisy counting and structured candidate exploration. Representative algorithms first mine frequent itemsets and then derive rules, such as DiffFIM [35], PrivBasis [8], PrivSuper [36], and DPARM [10], while others mine associations more directly (e.g., HCRMine [37]). These approaches often provide strong utility when a trusted curator is acceptable, but are less suitable in settings where raw data cannot be centralized.

Local DP for transactional mining: Local differential privacy (LDP) [14] perturbs data on the user side, eliminating the need for a trusted curator but typically introducing larger noise and heavier protocols for transactional mining. Representative protocols include LDPMiner [23] for heavy hitters/frequent items and SVIM/SVSM [12] for frequent items and itemsets; Priv_OA [22] targets small datasets with high accuracy but incurs substantial communication cost. In general, purely localized transactional mining can suffer from high runtime/communication overhead and sensitivity to data-dependent protocol parameters.

Hybrid and decentralized paradigms: Hybrid designs such as BLENDER [17] and Calibrate [38] combine local perturbation with curator-side post-processing to improve accuracy for popular record or heavy-hitter tasks. However, extending such pipelines to FIM/ARM remains challenging due to combinatorial candidate growth and repeated estimation over large itemset domains. Federated learning provides another decentralized trajectory [39], [40], and recent studies explore FIM in this paradigm [41].

Positioning of CM-ARM: CM-ARM is closest to LDP-based transactional mining and hybrid local–central designs, and it addresses their practical bottlenecks by centralizing only coordination-critical steps (candidate selection and private truncation-length determination) while keeping frequency reporting localized, thereby reducing overhead and improving robustness to dataset-dependent reporting settings.

VIII. Concluding Remarks

This paper presents CM-ARM, a privacy-preserving association rule mining framework that combines centralized candidate selection and private truncation-length determination with localized reporting for frequent itemset and rule discovery. This hybrid design reduces the overhead of purely localized approaches while avoiding full reliance on a trusted curator.

Experimental results show that, under the same privacy budgets, CM-ARM improves both runtime and utility over localized and centralized baselines. In particular, it achieves up to $563\times$ end-to-end speedup and yields higher utility measured by Hits, F-score, NCR, and support estimation error (see Section VI). Overall, this work advances association rule mining by offering a practical solution for privacy-preserving data mining in diverse settings.

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